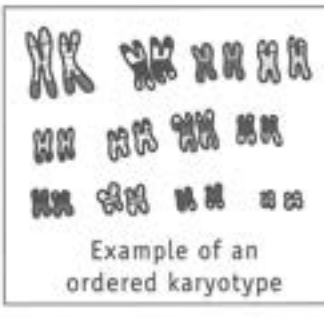
Karyotypes

Karyotype: a display of homologous chromosomes arranged in order of size and shape.

In order to study the chromosomes of an individual, geneticists often make karyotypes. To do this, chromosomes are extracted, stained and photographed. Each chromosome then shows a distinctive pattern of bands which are the key to pairing (shapes can vary slightly). The jumble of chromosomes must be sorted and paired — usually by entering the photograph into a computer and electronically rearranging it to illustrate the genetic make up of the individual being studied. The following exercise is a hypothetical 'cut and paste' version of this process.

1. Cut out the chromosomes below, using the dotted lines as a cutting guide.
2. Glue the chromosomes in pairs in your book to represent an ordered karyotype.



Example of an ordered karyotype





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